Lesson 14 Teacher's Key: Raising the Stakes

Scratch - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Replacing two hardcoded literals (a fixed spawn count, a fixed movement speed) with formulas keyed off the wave variable - the payoff lesson for simplifications flagged as deliberate back in Lesson 11.

Level 1 - Full Worked Solution

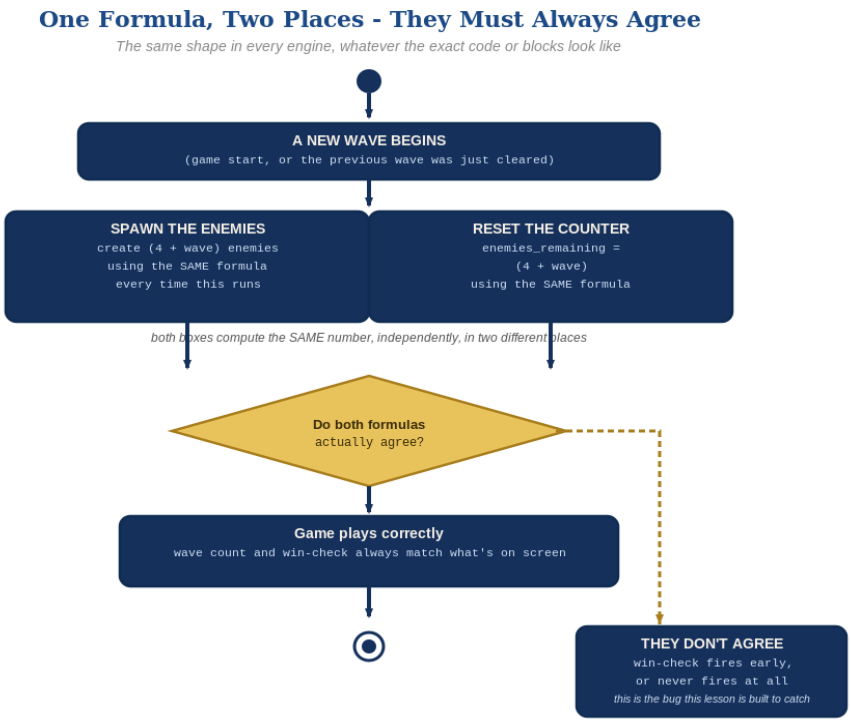
- "repeat (4 + wave)" and "set enemies_remaining to (4 + wave)" must use the identical expression - any mismatch between them is exactly the kind of bug this Key's troubleshooting section calls out.
- No new variables are needed for Level 1 - this is purely a formula change to numbers that were already there.
- Wave 1 still spawns exactly 5 (4 + 1 = 5), so nothing about the game's opening feel changes - only later waves escalate.

Highlighted lines are new this week; everything else is code you already have.

```
Enemy's "when I receive [spawn enemies]" script (revised):
when I receive [spawn enemies]
repeat (4 + wave)
  create clone of myself

GameManager's "when I receive [start game]" script (revised,
enemies_remaining line only):
set enemies_remaining to (4 + wave)

GameManager's "when I receive [enemy hit]" wave-advance (else) branch
(revised, enemies_remaining line only):
set enemies_remaining to (4 + wave)
```



Whether it's two places (most engines) or the loop and the reset living in genuinely different scripts (Godot, GameMaker), the rule is the same: one formula, referenced everywhere.

VIDEO GAME CREATION

Whether the spawn loop and the reset live in two places or two genuinely different scripts, the rule is the same: one formula, referenced everywhere.

Why This Structure

- Deriving both the spawn count and the enemies_remaining reset from the SAME formula (rather than one being "4 + wave" and the other a separately-updated literal) is the exact same discipline as Lesson 11's "broadcast [spawn enemies]" reuse - one source of truth, referenced everywhere it's needed.
- Starting the formula at "4 + wave" (rather than, say, "5 * wave") keeps wave 1 unchanged and growth gentle - worth discussing as a game-balance decision, not just a math one.

Level 2 / Level 3 Talking Points

- Ask students to compute the wave 1/2/3 enemy counts by hand from the formula before running it, then confirm the game matches.
- For strong students: discuss what would happen with a much steeper formula (e.g. "5 * wave") - wave 3 would have 15 enemies, likely overwhelming for a 3-wave game. A good real discussion about tuning difficulty, not just implementing it.

Level 2 - complete code:

Highlighted lines are the Level 2 additions on top of the Level 1 code above.

```
Enemy's "when I receive [spawn enemies]" script (Level 2 addition, on top
of the Level 1 code above):
when I receive [spawn enemies]
```

```
repeat (4 + wave)
  create clone of myself
set enemy_speed to (1 + (wave - 1) * 0.5)

Enemy clone's own forever loop (revised movement line only;
edge-check/escape logic unchanged):
forever
  change y by (enemy_speed * -1)
  // ...existing escape/edge-check logic unchanged...
```

Level 3 - no single combined code example:

This week's Level 3 is a "choose ONE" menu of five independent stretch features (destructible shields, a power-up, a bonus enemy, formation-marching movement, or a title/game-over screen) rather than the usual set of small, combinable stretch goals - so a single "all goals combined" code block doesn't fit Rule 14's usual model, and isn't provided here. Full worked solutions for all five options, across the active engines, are planned for the Lesson 15 wrap-up "Complete Codebase" send-home materials instead. Use the Talking Points above and the Handout's Level 3 menu to guide students toward whichever option they choose.

Common Student Mistakes

- Updating the spawn loop's count but forgetting to update enemies_remaining's reset value (or vice versa), causing the win-check to fire early or never.
- Using a different formula in two different places by accident (e.g. "4 + wave" in one, "5 + wave" in another).
- For Level 2: forgetting to recalculate enemy_speed inside the spawn-enemies script itself, so it only ever gets wave 1's value.